

Pascal Triangle Complete DSA Notes (Basic to Advanced)

1. What is Pascal Triangle?

Pascal Triangle is a triangular arrangement of numbers where every value is created by adding the two values above it.

Example:

```
1
1 1
1 2 1
1 3 3 1
1 4 6 4 1
```

Properties:

- First and last element of every row is 1
- Every internal element:

value = upper-left + upper-right

2. Mathematical Formula

Pascal Triangle is based on combinations.

Every element:

row = n
column = r

Value:

$$nC_r = n! / (r! * (n-r)!)$$

Example:

Row 4:

```
4C0 4C1 4C2 4C3 4C4
1 4 6 4 1
```

3. Important Properties

1. Row number n has $n+1$ elements.

2. Sum of n th row:

$$2^n$$

3. Symmetry:

$$nC_r = nC_{(n-r)}$$

4. Fibonacci numbers can be derived from diagonal sums.

5. Used heavily in:

- Combinatorics
- Probability
- Dynamic Programming
- Binomial Expansion

4. Binomial Expansion Connection

Pascal Triangle gives coefficients of:

$$(a+b)^n$$

Example:

$$(a+b)^3$$

Coefficients:

1 3 3 1

Expansion:

$$a^3 + 3a^2b + 3ab^2 + b^3$$

5. Generate Complete Pascal Triangle

Approach:

Create rows one by one.

For every index:

if first or last:
put 1

else:
previous row values sum

Python - Generate Pascal Triangle

```
def generate(numRows):  
    triangle=[]  
    for i in range(numRows):  
        row=[1]*(i+1)  
        for j in range(1,i):  
            row[j]=(  
                triangle[i-1][j-1]  
                +  
                triangle[i-1][j]  
            )  
        triangle.append(row)  
  
    return triangle
```

Time: $O(n^2)$
Space: $O(n^2)$

C++ - Generate Pascal Triangle

```
vector<vector<int>> generate(int n){  
    vector<vector<int>> ans;  
  
    for(int i=0;i<n;i++){  
        vector<int> row(i+1,1);  
  
        for(int j=1;j<i;j++){  
            row[j]=  
                ans[i-1][j-1]  
                +  
                ans[i-1][j];  
        }  
        ans.push_back(row);  
    }  
  
    return ans;  
}
```

Java - Generate Pascal Triangle

```
List<List<Integer>> generate(int n){
    List<List<Integer>> ans=new ArrayList<>();

    for(int i=0;i<n;i++){
        List<Integer> row=new ArrayList<>();

        for(int j=0;j<=i;j++)
            row.add(1);

        for(int j=1;j<i;j++){
            row.set(
                j,
                ans.get(i-1).get(j-1)
                +
                ans.get(i-1).get(j)
            );
        }

        ans.add(row);
    }

    return ans;
}
```

6. Find Single Element nCr Efficiently

Instead of factorials, calculate iteratively.

Avoids overflow.

Formula transformation:

```
answer *= (n-r+i)
answer /= i
```

Complexity:

$O(r)$

```
def nCr(n,r):
    ans=1

    for i in range(1,r+1):
        ans*=n-r+i
        ans//=i

    return ans
```

7. Generate Specific Row

Problem:
Return kth row only.

Optimization:

Space $O(k)$

```
def getRow(k):  
    row=[1]  
  
    for i in range(1,k+1):  
        row.append(  
            row[-1]*(k-i+1)//i  
        )  
  
    return row
```

8. Pascal Triangle Dynamic Programming View

State:

$dp[i][j]$

Transition:

```
dp[i][j] =  
dp[i-1][j-1]  
+  
dp[i-1][j]
```

This is one of the simplest 2D DP examples.

9. Advanced Applications

Pascal Triangle appears in:

1. Combinations
2. Probability
3. Counting paths
4. Binomial theorem
5. Dynamic Programming
6. Polynomial coefficients
7. Combinatorial optimization

Advanced Problems:

- Pascal Triangle I
- Pascal Triangle II
- Unique Paths
- Combination Sum concepts
- Binomial Coefficient DP

Interview Patterns

Three common interview questions:

1. Generate full triangle

Time:
 $O(n^2)$

2. Return kth row

Time:
 $O(k)$

3. Find element at row,column

Use:
 nCr

Time:
 $O(r)$